

HISTORY LOG

Back to editor

MINN2024A RESEARCH PROJECT

main.tex

Edited

Viewing 17th July - 11:29 am

10 changes in main.tex

1 \documentclass{article}

2 \usepackage{graphicx} % Required for inserting images

3

4 \title{Minn2024A

5 Research Project

6 BLASTSMART APP PROGRAMMING

7

8 Name: Sandiso (2557238)}

9 \author{Dr. Milka Madahanna

10 Mr. Tebatso Phala }

11 \date{July 2026}

12

13 \begin{document}

14

15 \maketitle

16

17 \section{Abstract}

18

19 \textbf{Introduction}

20

21 Dwala Global Resources is creating a mobile app called BlastSmart to help blasting engineers in South African mines. The app will make blast design calculations easier and faster than using paper books. The project has three phases, but this report will only focus on Phase 1, which is the Blast Calculator. The aim of this project is to build a simple mobile calculator that helps engineers do blast calculations quickly and correctly. The engineer will enter information such as bench height, burden, spacing, blast hole diameter, and other blast details. The app will then calculate the amount of explosive needed, the actual powder factor, the volume of rock, stemming length, and other important factors. The objectives are to make blast calculations easy, reduce mistakes, save time, and better blasting operations.

7 more updates below

All history

Labels

Yesterday

Edited main.tex

You

16th July, 1:17 pm

Edited main.tex

You

16th July, 12:49 pm

Edited main.tex

Sandiso Mabuza

16th July, 12:18 pm

Edited main.tex

You

Wed, 18th Feb 26

18th February, 4:58 pm

Created main.tex

Back to editor

MINN2024A RESEARCH PROJECT

main.tex

Edited

Viewing 17th July - 11:29 am

10 changes in main.tex

1 \documentclass{article}

2 \usepackage{graphicx} % Required for inserting images

3

4 \title{Minn2024A

5 Research Project

6 BLASTSMART APP PROGRAMMING

7

8 Name: Sandiso (2557238)}

9 \author{Dr. Milka Madahanna

10 Mr. Tebatso Phala }

11 \date{July 2026}

12

13 \begin{document}

14

15 \maketitle

16

17 \section{Abstract}

18

19 \textbf{Introduction}

20

21 Dwala Global Resources is creating a mobile app called BlastSmart to help blasting engineers in South African mines. The app will make blast design calculations easier and faster than using paper books. The project has three phases, but this report will only focus on Phase 1, which is the Blast Calculator. The aim of this project is to build a simple mobile calculator that helps engineers do blast calculations quickly and correctly. The engineer will enter information such as bench height, burden, spacing, blast hole diameter, and other blast details. The app will then calculate the amount of explosive needed, the actual powder factor, the volume of rock, stemming length, and other important factors. The objectives are to make blast calculations easy, reduce mistakes, save time, and better blasting operations.

7 more updates below

All history

Labels

Today

main.tex

Sandiso Mabuza

Yesterday

16th July, 3:10 pm

Edited main.tex

You

16th July, 2:43 pm

Edited main.tex

Sandiso Mabuza

16th July, 2:38 pm

Edited main.tex

Sandiso Mabuza

16th July, 1:33 pm

Edited main.tex

You

Back to editor

MINN2024A RESEARCH PROJECT

main.tex Edited

Viewing 17th July · 11:29 am

10 changes in main.tex

All history

Labels

1 \documentclass{article}

2 \usepackage{graphicx} % Required for inserting images

3

4 \title{Minn2024A

5 Research Project

6 BLASTSMART APP PROGRAMMING

7

8 Name: Sandiso (2557238)}

9 \author{Dr. Milka Madahanna

10 Mr. Tebatso Phala }

11 \date{July 2026}

12

13 \begin{document}

14

15 \maketitle

16

17 \section{Abstract}

18

19 \textbf{Introduction}

20

21 Dwaia Global Resources is creating a mobile app called BlastSmart to help blasting engineers in South African mines. The app will make blast design calculations easier and faster than using paper books. The project has three phases, but this report will only focus on Phase 1, which is the Blast Calculator. The aim of this project is to build a simple mobile calculator that helps engineers do blast calculations quickly and correctly. The engineer will enter information such as bench height, burden, spacing, blast hole diameter, and other blast details. The app will then calculate the amount of explosive needed, the actual powder factor, the volume of rock, stemming length, and other important parameters. The objectives are to make blast calculations easy, reduce mistakes, save time, and better blasting operations.

Today

17th July, 11:05 am

Edited

main.tex

You Sandiso Mabuza

17th July, 11:00 am

Edited

main.tex

Sandiso Mabuza

17th July, 10:55 am

Edited

main.tex

Sandiso Mabuza

17th July, 10:48 am

Edited

main.tex

Sandiso Mabuza

17th July, 10:43 am

Edited

main.tex

Sandiso Mabuza

7 more updates below

https://www.overleaf.com/project/6995d390242c1e1d145ae43

File Edit Insert View Help Upgrade

MINN2024A RESEARCH PROJECT

Recompile

2 / 5

74%

main.tex

Code Visual Viewing

34 \item The final program was able to calculate important blasting values such as burden, stemming length, explosive mass, actual powder factor, and mean fragment size.

35

36 \item The completed Python program will be used in the next stage of the BlastSmart mobile application so that blasting engineers can do calculations quickly and easily on their mobile devices.

37 \end{itemize}

38

39

40

41 \textbf{RESULTS}

42 \begin{itemize}

43

44 \textbf{3. ALGORITHMS}

45 }3.1) Algorithm for Surface Blasting Design Calculator and Flow Chart

46

47 \item Start the surface blasting calculator.

48 \item Enter the blasting diameter.

49 \item Enter the average in-hole density.

50 \item Enter the spacing-to-burden ratio.

51 \item Enter the bench height.

52 \item Enter whether the blasting location is close to a community or not.

53 \item Enter whether the rock type is hard or soft.

54 \item Calculate the mass of explosive.

55 \item Calculate the burden from the blasting diameter.

56 \item Calculate the spacing using the spacing-to-burden ratio and burden.

The final program was able to calculate important blasting values such as burden, stemming length, explosive mass, actual powder factor, and mean fragment size.

The completed Python program will be used in the next stage of the BlastSmart mobile application so that blasting engineers can do calculations quickly and easily on their mobile devices.

RESULTS

3. ALGORITHMS 3.1) Algorithm for Surface Blasting Design Calculator and Flow Chart 3.2) Algorithm for Fragmentation Calculator and Flow Chart

4.1. Surface Design Calculator BEGIN Surface Blasting Design Calculator INPUT Blasting Diameter INPUT Average In-Hole Density INPUT Spacing-To-Burden-Ratio INPUT Bench Height INPUT Proximity INPUT Rock-Type CONVERT Proximity TO UPPER CASE CONVERT Rock-Type TO UPPER CASE Burden = math.sqrt((Mass_of_explosive/(Bench height - Stemming length)))/(Spacing to burden ratio Technical powder factor) Mass of Explosive = (Average In-Hole Density (Blasting Diameter) (1273 Burden Spacing = Spacing-To-Burden-Ratio Burden) Proximity = "CLOSE TO COMMUNITY" THEN Stemming Length = (Blasting Diameter) ELSE IF Proximity = "NOT" THEN Stemming Length = 20 (Blasting Diameter) END IF Subdrill = Burden IF Rock-Type = "HARD" THEN Actual Powder Factor = (Bench Height - Stemming Length + Subdrill) (Burden Spacing Bench Height) ELSE IF Rock-Type = "SOFT" THEN Actual Powder Factor = (Bench Height - Stemming Length - Subdrill) (Burden Spacing Bench Height) END IF OUTPUT Burden OUTPUT Stemming Length OUTPUT Actual Powder Factor IF Next Step = "YES" THEN INPUT Bench Height INPUT Relative Effective Energy of Explosive INPUT Stemming Length 2 Mean Fragment Size = (Average In-Hole Density x (Bench Height - Stemming Length 2)) ÷ Relative Effective Energy of Explosive OUTPUT Mean Fragment Size, "mm" ELSE OUTPUT "Invalid" END IF END Fragmentation Calculator

5. CODE 4. PSEUDOCODE

4.1. Surface Design Calculator BEGIN Surface Blasting Design Calculator INPUT Blasting Diameter INPUT Average In-Hole Density INPUT Spacing-To-Burden-Ratio INPUT Bench Height INPUT Proximity INPUT Rock-Type CONVERT Proximity TO UPPER CASE CONVERT Rock-Type TO UPPER CASE Burden = math.sqrt((Mass_of_explosive/(Bench height - Stemming length)))/(Spacing to burden ratio Technical powder factor) Mass of Explosive = (Average In-Hole Density (Blasting Diameter) (1273 Burden Spacing = Spacing-To-Burden-Ratio Burden) Proximity = "CLOSE TO COMMUNITY" THEN Stemming Length = (Blasting Diameter) ELSE IF Proximity = "NOT" THEN Stemming Length =

main.tex Edited

Viewing 17th July · 1:44 pm

101 changes in main.tex

All history

Labels

70 Ask the user whether the mean fragment size must be calculated.

71 When the answer is "NO", display "Thank You".

72 When the answer is "YES", enter the bench height.

73 Enter the relative effective energy of the explosive.

74 Enter the stemming length.

75 Use the average in-hole density entered in the surface blasting calculator.

76 Calculate the mean fragment size.

77 Display the mean fragment size in millimetres.

78 When the user enters an option other than "YES" or "NO", display "Invalid".

79 End the program.

80 \item

81

82 \end{itemize}

83

84

85

86 4. \textbf{PSEUDOCODE}

87 \begin{itemize}

88

89 \textbf{4.1. \textbf{Surface Design Calculator}}

90 }

91

92 BEGIN Surface Blasting Design Calculator

93 INPUT Blasting Diameter

94 INPUT Average In-Hole Density

95 INPUT Spacing-To-Burden-Ratio

96 INPUT Bench Height INPUT Proximity

97 INPUT Rock-Type

98 CONVERT Proximity TO UPPER CASE

99 CONVERT Rock-Type TO UPPER CASE

Today

17th July, 12:55 pm

Edited

main.tex

Kamogelo Maditsi Sandiso Mabuza

17th July, 12:48 pm

Edited

main.tex

Sandiso Mabuza

17th July, 12:43 pm

Edited

main.tex

Sandiso Mabuza

17th July, 12:36 pm

Edited

main.tex

Sandiso Mabuza You

17th July, 12:31 pm

Edited

main.tex

Sandiso Mabuza You

72 more updates below

